Grade 11 – Term 1
C5 Practice Activity 5 with Solutions

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Evaluated criterion

C5. Calculates and interprets the mean and variance of continuous distributions using standard formulas and functions.

1 Introduction

Problems:

1.1. Purpose and learning goals

1.2. Continuous probability distributions

1.3. Mean, variance, and standard deviation

1.4. Formula reference

Criterion C4 focuses on calculating and interpreting measures that describe the center and spread of a continuous probability distribution. The main measures used are the mean, or expected value, the variance, and the standard deviation.

These measures summarize different features of a probability distribution. The mean identifies its center, while the variance and standard deviation describe how widely values are distributed around that center. The appropriate formulas depend on the probability distribution and its parameters.

1.1 Purpose and learning goals

The purpose of this activity is to calculate and interpret the mean, variance, and standard deviation of continuous probability distributions using standard formulas.

Measure	Common symbols	Meaning
Mean	$E(X)$, $\bar{x}$, μ	Expected or average value
Variance	$Var(X)$, s^2, σ^2	Average squared spread from the mean
Standard deviation	$SD(X)$, s, σ	Typical spread from the mean

For each situation:

1. Identify the probability distribution and its parameters.
2. Calculate the mean, or expected value, $E(X)$.
3. Calculate the variance, $Var(X)$.
4. Calculate the standard deviation, $\sigma_X = \sqrt{Var(X)}$.
5. Interpret the results in the context of the variable.



1.2 Continuous probability distributions

The distributions used in this criterion describe continuous variables. Their probability density functions assign probability to intervals through area under the curve. Different distributions have different shapes and parameters, so their means and variances are calculated using different formulas.

The distributions used in this activity are:

- Uniform
- Triangular
- Linear increasing
- Linear decreasing
- Trapezoidal
- Exponential
- Normal

The notation identifies the distribution and its parameters. For example,

$$X \sim U(a, b)$$

indicates a uniform distribution between a and b, while

$$X \sim N(\mu, \sigma^2)$$

indicates a normal distribution with mean μ and variance σ^2.



1.3 Mean, variance, and standard deviation

The mean, written as $E(X)$, describes the center of a probability distribution. For a continuous random variable, it represents the long-run average value expected from repeated observations under the model.

Variance measures the spread of the distribution around its mean:

$$Var(X) = E\left[(X - E(X))^2\right].$$

The variance is expressed in squared units. Its square root is the standard deviation:

$$\sigma_X = \sqrt{Var(X)}.$$

Unlike variance, the standard deviation has the same units as the original variable, which makes it easier to interpret in context.

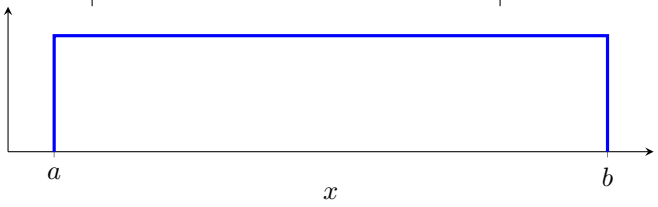
For every problem, first identify the distribution and its parameters. Then select the corresponding formulas, substitute the parameters, calculate the results, and interpret their meaning using the units of the variable.

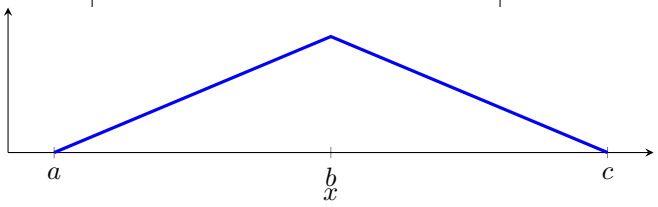


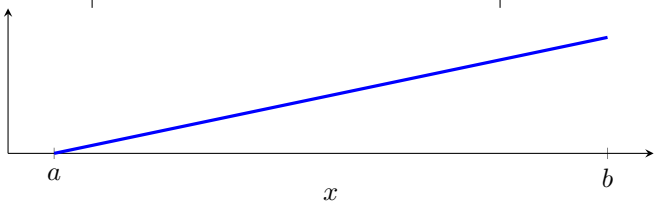
1.4 Formula reference

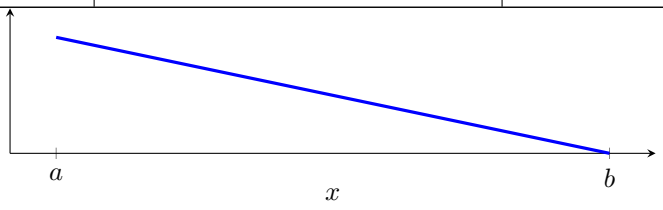
The following table summarizes the formulas used in this criterion.

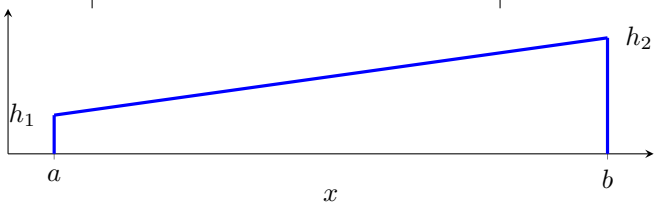
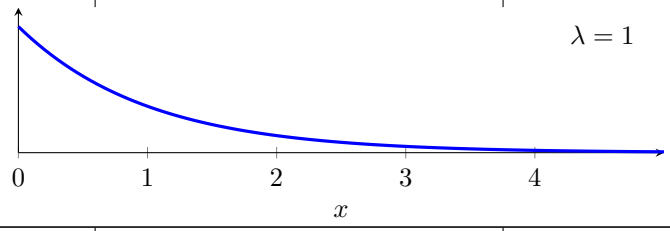
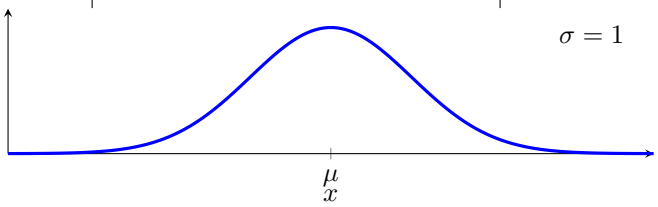
Distribution	Notation	$E(X)$	$Var(X)$
Uniform	$X \sim U(a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Triangular	$X \sim \text{Tri}(a, b, c)$	$\frac{a+b+c}{3}$	$\frac{a^2+b^2+c^2-ab-ac-bc}{18}$
Linear increasing	$X \sim \text{Lin}_{\text{inc}}(a, b)$	$\frac{a+2b}{3}$	$\frac{(b-a)^2}{18}$
Linear decreasing	$X \sim \text{Lin}_{\text{dec}}(a, b)$	$\frac{2a+b}{3}$	$\frac{(b-a)^2}{18}$







Distribution	Notation	$E(X)$	$Var(X)$
			

Distribution	Notation	$E(X)$	$Var(X)$
Trapezoidal	$X \sim \text{Trap}(a, b, h_1, h_2)$	$\frac{b-a}{3}\frac{h_1^2+h_1h_2+h_2^2}{h_1+h_2}+\frac{a+b}{2}$	$\frac{(b-a)^2}{18}\frac{h_1^2+2h_1h_2+h_2^2}{(h_1+h_2)^2}$
			
Exponential	$X \sim \text{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
			
Normal	$X \sim N(\mu, \sigma^2)$	μ	σ^2
			

The standard deviation is obtained after calculating the variance:

$$\sigma_X = \sqrt{Var(X)}.$$

When interpreting a result, report the mean and standard deviation using the units of the original variable, while variance uses squared units.

The formulas above should be applied directly once the distribution and its parameters have been identified. The second table provides a compact reference without the graphs.

Distribution	Notation	$E(X)$	$Var(X)$
Uniform	$X \sim U(a, b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Triangular	$X \sim \text{Tri}(a, b, c)$	$\frac{a+b+c}{3}$	$\frac{a^2+b^2+c^2-ab-ac-bc}{18}$
Linear increasing	$X \sim \text{Lin}_{\text{inc}}(a, b)$	$\frac{a+2b}{3}$	$\frac{(b-a)^2}{18}$

Distribution	Notation	$E(X)$	$Var(X)$
Linear decreasing	$X \sim \text{Lin}_{\text{dec}}(a, b)$	$\frac{2a+b}{3}$	$\frac{(b-a)^2}{18}$
Trapezoidal	$X \sim \text{Trap}(a, b, h_1, h_2)$	$\frac{b-a}{3}\frac{h_1^2+h_1h_2+h_2^2}{h_1+h_2}+\frac{a+b}{2}$	$\frac{(b-a)^2}{18}\frac{h_1^2+2h_1h_2+h_2^2}{(h_1+h_2)^2}$
Exponential	$X \sim \text{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	$X \sim N(\mu, \sigma^2)$	μ	σ^2



2 Introductory Problems

Problems:

2.1. Uniform distribution

2.2. Linear increasing distribution

2.3. Linear decreasing distribution

2.4. Triangular distribution

2.5. Exponential distribution

2.6. Normal distribution

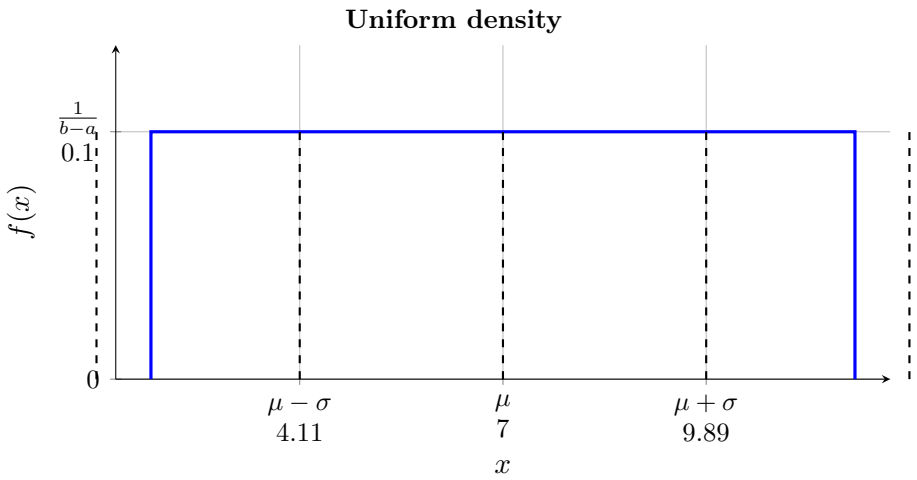
2.1 Uniform distribution

Distribution and parameters

A small business invests in a short-term opportunity. Based on similar investments, the money earned is expected to be between \$2,000 and \$12,000, with all values in this range considered equally likely.

Let X represent the money earned, measured in thousands of dollars. Therefore,

$$X \sim U(a,b) \quad \longrightarrow \quad X \sim U(2,12) \quad \longrightarrow \quad a = 2, \quad b = 12.$$



Mean

For a uniform distribution,

$$E(X) = \frac{a+b}{2} \quad \longrightarrow \quad E(X) = \frac{2+12}{2} = 7 \quad \longrightarrow \quad E(X) = 7 \text{ thousand dollars}$$

Variance

For a uniform distribution,

$$Var(X) = \frac{(b-a)^2}{12} \qquad \longrightarrow \qquad Var(X) = \frac{(12-2)^2}{12} = \frac{100}{12} = \frac{25}{3} \approx 8.33 \qquad \longrightarrow \qquad \boxed{Var(X) \approx 8.33 \text{ thousand dollars}^2}$$

Standard Deviation

The standard deviation is

$$\sigma_X = \sqrt{\frac{25}{3}} \approx 2.89 \qquad \longrightarrow \qquad \boxed{\sigma_X \approx 2.89 \text{ thousand dollars}}$$

Probability within one standard deviation

We want the probability that the money earned is within one standard deviation of the mean:

$$P(\mu - \sigma_X < X < \mu + \sigma_X)$$

Using $\mu = 7$ and $\sigma_X \approx 2.89$,

$$P(7 - 2.89 < X < 7 + 2.89) = P(4.11 < X < 9.89)$$

For a uniform distribution,

$$P(c < X < d) = \frac{d-c}{b-a}$$

Therefore,

$$P(4.11 < X < 9.89) = \frac{9.89 - 4.11}{12 - 2} = \frac{5.78}{10} \approx 0.578$$

$$\boxed{P(\mu - \sigma_X < X < \mu + \sigma_X) \approx 0.578}$$

Thus, approximately 57.8% of the possible investment earnings are within one standard deviation of the mean.

Probability within one standard deviation for any uniform distribution

For a uniform distribution $X \sim U(a, b)$, the mean and standard deviation are

$$\mu = \frac{a+b}{2} \qquad \text{and} \qquad \sigma_X = \frac{b-a}{\sqrt{12}}.$$

The interval within one standard deviation of the mean is

$$\mu - \sigma_X < X < \mu + \sigma_X.$$

Its total width is

$$(\mu + \sigma_X) - (\mu - \sigma_X) = 2\sigma_X = \frac{2(b-a)}{\sqrt{12}}.$$

Since the probability of an interval in a uniform distribution is its width divided by the total width $b - a$,

$$P(\mu - \sigma_X < X < \mu + \sigma_X) = \frac{2\sigma_X}{b-a} = \frac{2}{\sqrt{12}} = \frac{1}{\sqrt{3}} \approx 0.577.$$

Therefore, for every uniform distribution,

$$P(\mu - \sigma_X < X < \mu + \sigma_X) = \frac{1}{\sqrt{3}} \approx 57.7\%$$

Thus, the probability within one standard deviation of the mean is the same for every continuous uniform distribution.

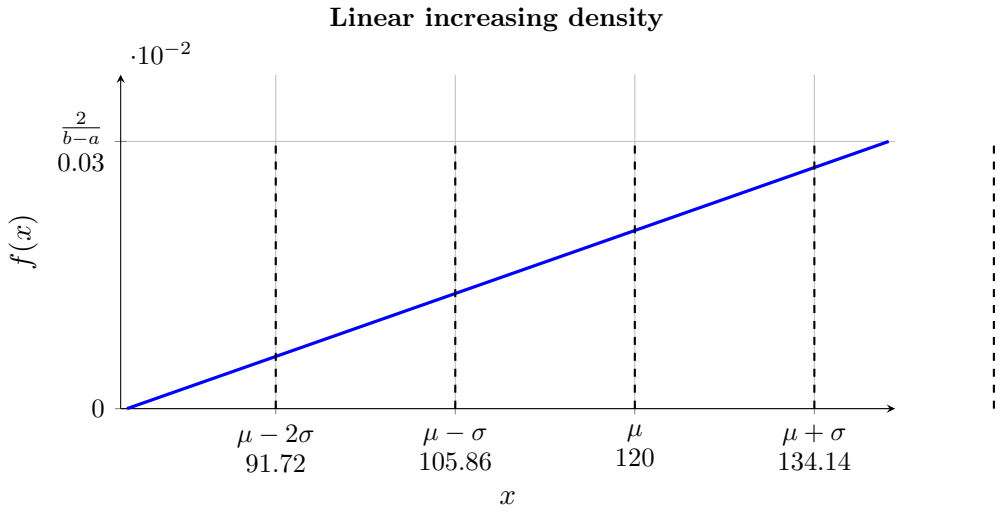
2.2 Linear increasing distribution

Distribution and parameters

A bakery prepares custom cakes for weekend orders. Based on previous weekends, the preparation time for a randomly selected cake ranges from 80 to 140 minutes. Longer preparation times are more likely because larger and more detailed cakes require more work, so the probability density increases linearly across this range.

Let T represent the preparation time, in minutes, for one cake. Therefore,

$$T \sim \text{Lin}_{\text{inc}}(a, b) \quad \longrightarrow \quad T \sim \text{Lin}_{\text{inc}}(80, 140) \quad \longrightarrow \quad a = 80 \quad b = 140$$



Mean

For a linear increasing distribution,

$$E(T) = \frac{a + 2b}{3} \quad \longrightarrow \quad E(T) = \frac{80 + 2(140)}{3} = \frac{360}{3} = 120 \quad \longrightarrow \quad \boxed{E(T) = 120 \text{ minutes}}$$

Variance

For a linear increasing distribution,

$$Var(T) = \frac{(b-a)^2}{18} \quad \longrightarrow \quad Var(T) = \frac{(140-80)^2}{18} = \frac{3600}{18} = 200 \quad \longrightarrow \quad \boxed{Var(T) = 200 \text{ minutes}^2}$$

Interpretation

The standard deviation is

$$\sigma_T = \sqrt{200} \approx 14.14 \qquad \longrightarrow \qquad \boxed{\sigma_T \approx 14.14 \text{ minutes}}$$

Probability within one standard deviation

We want the probability that the preparation time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using $\mu = 120$ and $\sigma_T \approx 14.14$,

$$P(120 - 14.14 < T < 120 + 14.14) = P(105.86 < T < 134.14)$$

For a linear increasing distribution, the probability density is

$$f(t) = \frac{2(t-a)}{(b-a)^2}. \qquad \longrightarrow$$

$$\begin{aligned} P(105.86 < T < 134.14) &= \int_{105.86}^{134.14} \frac{2(t-80)}{(140-80)^2}\, dt \\ &= \left[\frac{(t-80)^2}{3600}\right]_{105.86}^{134.14} \\ &\approx 0.6285 \end{aligned}$$

$$\boxed{P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.6285}$$

Thus, approximately 62.84% of cake preparation times are within one standard deviation of the mean.

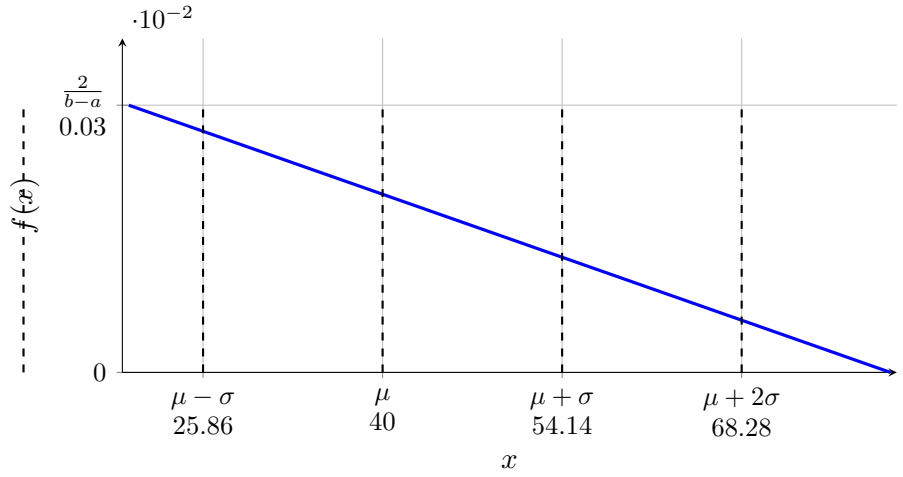
2.3 Linear decreasing distribution

Distribution and parameters

A bakery delivers custom cake orders to customers during the weekend. Based on previous deliveries, the delivery time for a randomly selected order ranges from 20 to 80 minutes. Shorter delivery times are more likely because most orders are delivered within the local area, so the probability density decreases linearly as the delivery time increases.

Let T represent the delivery time, in minutes, for one order. Therefore,

$$T \sim \text{Lin}_{\text{dec}}(a, b) \qquad \longrightarrow \qquad T \sim \text{Lin}_{\text{dec}}(20, 80) \qquad \longrightarrow \qquad a = 20, \qquad b = 80$$



Mean

For a linear decreasing distribution,

$$E(T) = \frac{2a+b}{3} \quad \longrightarrow \quad E(T) = \frac{2(20)+80}{3} = \frac{120}{3} = 40 \quad \longrightarrow \quad \boxed{E(T) = 40 \text{ minutes}}$$

Variance

For a linear decreasing distribution,

$$Var(T) = \frac{(b-a)^2}{18} \quad \longrightarrow \quad Var(T) = \frac{(80-20)^2}{18} = \frac{3600}{18} = 200 \quad \longrightarrow \quad \boxed{Var(T) = 200 \text{ minutes}^2}$$

Interpretation

The standard deviation is

$$\sigma_T = \sqrt{200} \approx 14.14 \quad \longrightarrow \quad \boxed{\sigma_T \approx 14.14 \text{ minutes}}$$

Probability within one standard deviation

We want the probability that the delivery time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using $\mu = 40$ and $\sigma_T \approx 14.14$,

$$P(40 - 14.14 < T < 40 + 14.14) = P(25.86 < T < 54.14)$$

For a linear decreasing distribution, the probability density is

$$f(t) = \frac{2(b-t)}{(b-a)^2}. \quad \longrightarrow$$

$$P(25.86 < T < 54.14) = \int_{25.86}^{54.14} \frac{2(80-t)}{(80-20)^2}\, dt$$

$$= \left[\frac{2(80)t - t^2}{3600}\right]_{25.86}^{54.14}$$

$$\approx 0.6285$$

$$P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.6285$$

Thus, approximately 62.84% of delivery times are within one standard deviation of the mean.

Probability within one standard deviation

The probability of obtaining a value within one standard deviation of the mean is the same for every linear increasing or decreasing distribution, regardless of the values of a and b.

For a linear increasing distribution,

$$f(x) = \frac{2(x-a)}{(b-a)^2}, \qquad \mu = \frac{a+2b}{3}, \qquad \sigma = \frac{b-a}{\sqrt{18}}, \qquad a \le x \le b.$$

Let $d = b - a$. Then

$$\mu - a = \frac{2d}{3}, \qquad \sigma = \frac{d}{\sqrt{18}},$$

so

$$\mu - \sigma - a = d\left(\frac{2}{3} - \frac{1}{\sqrt{18}}\right), \qquad \mu + \sigma - a = d\left(\frac{2}{3} + \frac{1}{\sqrt{18}}\right).$$

Therefore,

$$\begin{aligned}
P(\mu - \sigma < X < \mu + \sigma) &= \int_{\mu-\sigma}^{\mu+\sigma} \frac{2(x-a)}{d^2}\, dx \\
&= \left[\frac{(x-a)^2}{d^2}\right]_{\mu-\sigma}^{\mu+\sigma} \\
&= \frac{(\mu+\sigma-a)^2 - (\mu-\sigma-a)^2}{d^2} \\
&= \left(\frac{2}{3} + \frac{1}{\sqrt{18}}\right)^2 - \left(\frac{2}{3} - \frac{1}{\sqrt{18}}\right)^2 \\
&= \left[\left(\frac{2}{3}\right)^2 + 2\left(\frac{2}{3}\right)\frac{1}{\sqrt{18}} + \frac{1}{18}\right] - \left[\left(\frac{2}{3}\right)^2 - 2\left(\frac{2}{3}\right)\frac{1}{\sqrt{18}} + \frac{1}{18}\right] \\
&= 4\left(\frac{2}{3}\right)\left(\frac{1}{\sqrt{18}}\right) \\
&= \frac{8}{3\sqrt{18}} = \frac{8}{9\sqrt{2}} \approx 0.6285.
\end{aligned}$$

For a linear decreasing distribution, the density is the reflection of the increasing density about the midpoint of $[a, b]$. Reflection preserves probabilities and standardizes the distribution in the same way. Hence,

$$P(\mu - \sigma < X < \mu + \sigma) = \frac{8}{3\sqrt{18}} \approx 62.85\%.$$

Thus, for every linear increasing or decreasing distribution on $[a, b]$,

$$P(\mu - \sigma < X < \mu + \sigma) = \frac{8}{3\sqrt{18}} = \frac{8}{9\sqrt{2}} \approx 62.85\%$$

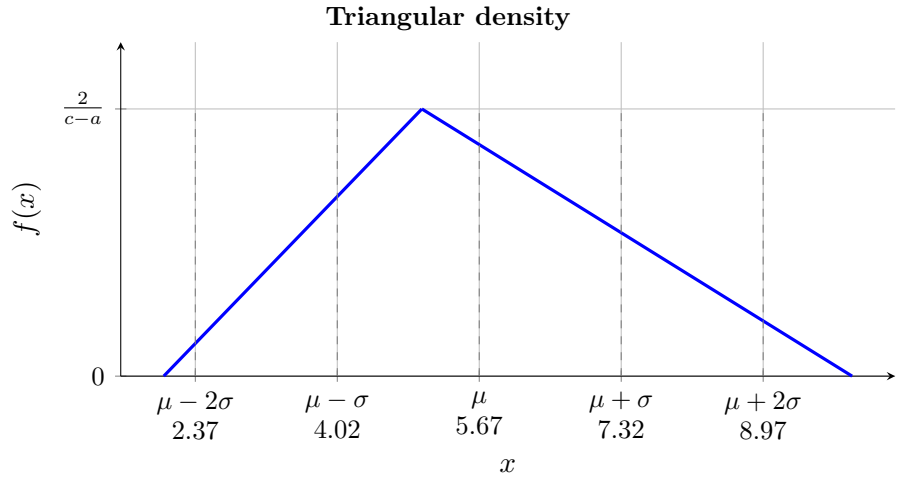
2.4 Triangular distribution

Distribution and parameters

A financial institution processes applications for small business loans. Based on previous applications, the processing time ranges from 2 to 10 hours. Most applications can be completed in about 5 hours, while shorter and longer processing times are less common. Therefore, the processing time has a triangular probability density with a mode at 5 hours.

Let T represent the processing time of a loan application, measured in hours. Therefore,

$$T \sim \text{Tri}(a, b, c) \quad \longrightarrow \quad T \sim \text{Tri}(2, 10, 5) \quad \longrightarrow \quad a = 2 \quad b = 10 \quad c = 5$$



Mean

For a triangular distribution,

$$E(T) = \frac{a+b+c}{3} \quad \longrightarrow \quad E(T) = \frac{2+10+5}{3} = \frac{17}{3} \approx 5.67 \quad \longrightarrow \quad E(T) \approx 5.67 \text{ hours}$$

Variance

For a triangular distribution,

$$Var(T) = \frac{a^2+b^2+c^2-ab-ac-bc}{18}.$$

$$\longrightarrow \quad \begin{aligned} Var(T) &= \frac{2^2+10^2+5^2-(2)(10)-(2)(5)-(10)(5)}{18} \\ &= \frac{4+100+25-20-10-50}{18} \\ &= \frac{49}{18} \approx 2.72 \end{aligned} \quad \longrightarrow \quad Var(T) \approx 2.72 \text{ hours}^2$$

Interpretation

The standard deviation is

$$\sigma_T = \sqrt{\frac{49}{18}} \approx 1.65 \qquad \longrightarrow \qquad \boxed{\sigma_T \approx 1.65 \text{ hours}}.$$

Probability within one standard deviation

We want the probability that the processing time is within one standard deviation of the mean:

$$P(\mu - \sigma_T < T < \mu + \sigma_T)$$

Using $\mu \approx 5.67$ and $\sigma_T \approx 1.65$,

$$P(5.67 - 1.65 < T < 5.67 + 1.65) = P(4.02 < T < 7.32)$$

Since the mode is $c = 5$, the probability density is

$$f(t) = \begin{cases} \dfrac{2(t-a)}{(b-a)(c-a)}, & a \le t \le c, \\ \dfrac{2(b-t)}{(b-a)(b-c)}, & c < t \le b. \end{cases}$$

Therefore,

$$P(4.02 < T < 7.32) = \int_{4.02}^{5} \frac{2(t-2)}{(10-2)(5-2)}\,dt + \int_{5}^{7.32} \frac{2(10-t)}{(10-2)(10-5)}\,dt$$

$$\approx 0.563$$

$$\boxed{P(\mu - \sigma_T < T < \mu + \sigma_T) \approx 0.563}$$

Thus, approximately 56.3% of loan-processing times are within one standard deviation of the mean.

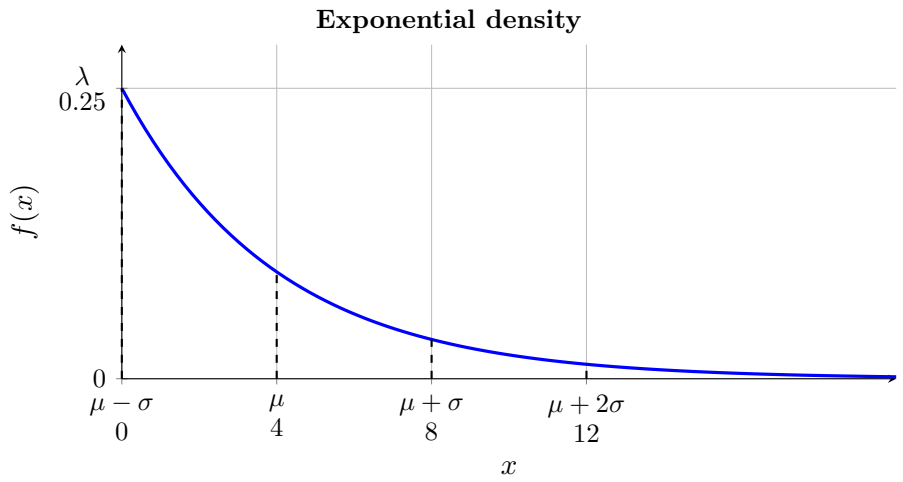
2.5 Exponential distribution

Parameter

An online clothing store receives orders throughout the day. Based on its recent sales activity, the store receives an average of 15 orders per hour. Let X represent the time between consecutive orders, measured in minutes.

For an exponential distribution, the rate is the average number of events per unit of time. Since 15 orders occur per hour,

$$\lambda = \frac{15}{60} = \frac{1}{4} \text{ min}^{-1} \qquad \longrightarrow \qquad \boxed{\lambda = \frac{1}{4} \text{ min}^{-1}}$$



Mean

For an exponential distribution,

$$E(X) = \frac{1}{\lambda} \quad \longrightarrow \quad E(X) = \frac{1}{1/4} = 4 \quad \longrightarrow \quad \boxed{E(X) = 4 \text{ minutes}}$$

Variance

For an exponential distribution,

$$Var(X) = \frac{1}{\lambda^2} \quad \longrightarrow \quad Var(X) = \frac{1}{(1/4)^2} = 16 \quad \longrightarrow \quad \boxed{Var(X) = 16 \text{ minutes}^2}$$

Interpretation

The standard deviation is

$$\sigma_X = \sqrt{16} = 4 \quad \longrightarrow \quad \boxed{\sigma_X = 4 \text{ minutes}}$$

The mean and standard deviation are both 4 minutes, reflecting the characteristic variability of an exponential waiting-time distribution.

Probability within one standard deviation

We want the probability that the time between consecutive orders is within one standard deviation of the mean:

$$P(\mu - \sigma_X < X < \mu + \sigma_X)$$

Using $\mu = 4$ and $\sigma_X = 4$,

$$P(4 - 4 < X < 4 + 4) = P(0 < X < 8)$$

For an exponential distribution, the cumulative distribution function is

$$P(X < x) = 1 - e^{-\lambda x}.$$

Therefore,

$$P(0 < X < 8) = 1 - e^{-(1/4)(8)} = 1 - e^{-2} \approx 0.8647$$

$$P(\mu - \sigma_X < X < \mu + \sigma_X) \approx 0.865$$

Thus, approximately 86.5% of the time intervals between consecutive orders are within one standard deviation of the mean.

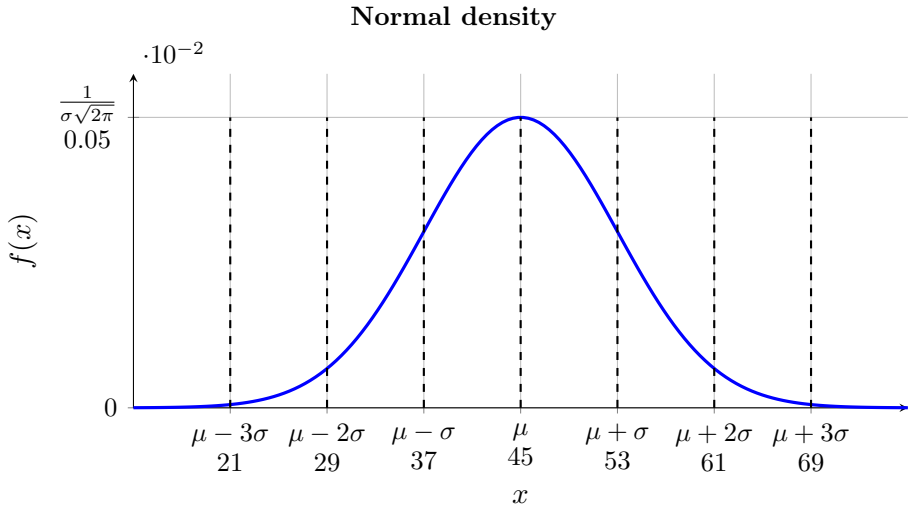
2.6 Normal distribution

Parameters

A small retail business tracks its monthly sales revenue over several months. Based on its historical sales data, the monthly revenue is approximately centered at \$45,000, with a standard deviation of \$8,000. Let Y represent the monthly sales revenue, measured in thousands of dollars.

Therefore,

$$Y \sim N(45, 8^2) \qquad \longrightarrow \qquad \mu = 45 \qquad \sigma = 8$$



Mean

For a normal distribution,

$$E(Y) = \mu \qquad \longrightarrow \qquad E(Y) = 45 \text{ thousand dollars}$$

Variance

For a normal distribution,

$$Var(Y) = \sigma^2 \qquad \longrightarrow \qquad Var(Y) = 8^2 = 64 \qquad \longrightarrow \qquad Var(Y) = 64 \text{ thousand dollars}^2 .$$

Interpretation

The standard deviation is

$$\sigma_Y = \sqrt{64} = 8 \qquad \longrightarrow \qquad \boxed{\sigma_Y = 8 \text{ thousand dollars}}.$$

This means that monthly sales revenue typically varies by about \$8,000 around the mean revenue of \$45,000.

Probability within one standard deviation

We want the probability that monthly sales revenue is within one standard deviation of the mean:

$$P(\mu - \sigma_Y < Y < \mu + \sigma_Y)$$

Using $\mu = 45$ and $\sigma_Y = 8$,

$$P(45 - 8 < Y < 45 + 8) = P(37 < Y < 53)$$

Standardizing the two endpoints,

$$z_1 = \frac{37 - 45}{8} = -1, \qquad z_2 = \frac{53 - 45}{8} = 1$$

Therefore,

$$P(37 < Y < 53) = P(-1 < Z < 1) \approx 0.6827$$

$$\boxed{P(\mu - \sigma_Y < Y < \mu + \sigma_Y) \approx 0.683}$$

Thus, approximately 68.3% of monthly sales revenues are within one standard deviation of the mean.

3 Context Problems

In each problem, the distribution is not named. Use the contextual clues to identify the underlying distribution and its parameters. Then calculate the mean, variance, and standard deviation, and interpret the mean and standard deviation in context.

Problems:

3.1. Airport shuttle arrival

3.2. Irrigation duration

3.3. Craft project time

3.4. Online auction price

3.5. Technical support time

3.6. Battery replacement age

3.7. Bicycle repair time

3.8. Truck loading time

3.9. Checkout arrivals

3.10. Machine breakdowns

3.11. Greenhouse temperature

3.12. Package filling mass

3.1 Airport shuttle arrival

An airport shuttle may arrive at any time from 8 to 20 minutes after a passenger reaches the stop, and every time in that interval is equally likely. Let X be the waiting time in minutes.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

The phrase "every time in that interval is equally likely" identifies $X \sim U(8, 20)$. Therefore,

$$E(X) = \frac{8+20}{2} = 14, \qquad Var(X) = \frac{(20-8)^2}{12} = 12, \qquad \sigma_X = \sqrt{12} \approx 3.46.$$

The average wait is 14 minutes, with a typical distance from that average of about 3.46 minutes.

3.2 Irrigation duration

A field's automatic irrigation cycle lasts between 30 and 54 minutes. A sensor selects the stopping time so that every duration in this interval is equally likely. Let X be the cycle duration in minutes.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

Equal likelihood throughout a fixed interval identifies $X \sim U(30, 54)$. Thus,

$$E(X) = 42, \qquad Var(X) = \frac{24^2}{12} = 48, \qquad \sigma_X = \sqrt{48} \approx 6.93.$$

Cycles average 42 minutes and typically differ from that average by about 6.93 minutes.

3.3 Craft project time

A beginner needs between 2 and 8 hours to finish a craft project. Longer times are progressively more likely, and the probability density rises in a straight line from zero at 2 hours to its maximum at 8 hours. Let X be the completion time in hours.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

A density that rises linearly from the lower endpoint identifies $X \sim \text{Lin}_{\text{inc}}(2, 8)$. Hence,

$$E(X) = \frac{2 + 2(8)}{3} = 6, \qquad Var(X) = \frac{(8-2)^2}{18} = 2, \qquad \sigma_X = \sqrt{2} \approx 1.41.$$

The mean completion time is 6 hours, with a standard deviation of about 1.41 hours.

3.4 Online auction price

The final price of a collectible lies between \$10 and \$25. Within this range, higher prices become progressively more likely, and the density increases in a straight line from zero at \$10 to its maximum at \$25. Let X be the price in dollars.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

The contextual shape identifies $X \sim \text{Lin}_{\text{inc}}(10, 25)$. Therefore,

$$E(X) = \frac{10 + 2(25)}{3} = 20, \qquad Var(X) = \frac{15^2}{18} = 12.5, \qquad \sigma_X = \sqrt{12.5} \approx 3.54.$$

The mean final price is \$20, with a standard deviation of about \$3.54.

3.5 Technical support time

A simple technical-support call lasts between 0 and 30 minutes. Very short calls are most likely, and the probability density decreases in a straight line to zero at 30 minutes. Let X be the call duration in minutes.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

A density that falls linearly to the upper endpoint identifies $X \sim \text{Lin}_{\text{dec}}(0, 30)$. Thus,

$$E(X) = \frac{2(0) + 30}{3} = 10, \qquad Var(X) = \frac{30^2}{18} = 50, \qquad \sigma_X = \sqrt{50} \approx 7.07.$$

Calls average 10 minutes and have a standard deviation of about 7.07 minutes.

3.6 Battery replacement age

A fleet replaces a certain battery between 12 and 48 months after installation. Replacement is most likely near 12 months, and the density decreases linearly to zero at 48 months. Let X be the replacement age in months.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

These clues identify $X \sim \text{Lin}_{\text{dec}}(12, 48)$. Hence,

$$E(X) = \frac{2(12) + 48}{3} = 24, \qquad Var(X) = \frac{36^2}{18} = 72, \qquad \sigma_X = \sqrt{72} \approx 8.49.$$

The mean replacement age is 24 months, with a standard deviation of about 8.49 months.

3.7 Bicycle repair time

A bicycle repair takes between 20 and 50 minutes. Repairs near 32 minutes are most common, and times become linearly less likely on either side of that peak. Let X be the repair time in minutes.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

A bounded density with one peak and straight sides identifies $X \sim \text{Tri}(20, 50, 32)$. Therefore,

$$E(X) = \frac{20 + 50 + 32}{3} = 34,$$

$$Var(X) = \frac{20^2 + 50^2 + 32^2 - (20)(50) - (20)(32) - (50)(32)}{18} = 38, \qquad \sigma_X = \sqrt{38} \approx 6.16.$$

Repairs average 34 minutes and typically differ from that average by about 6.16 minutes.

3.8 Truck loading time

Loading a delivery truck takes between 5 and 17 minutes. The most likely time is 8 minutes, and the density changes linearly from each endpoint to that peak. Let X be the loading time in minutes.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

The minimum, maximum, and single linearly connected peak identify $X \sim \text{Tri}(5, 17, 8)$. Thus,

$$E(X) = \frac{5 + 17 + 8}{3} = 10,$$

$$Var(X) = \frac{5^2 + 17^2 + 8^2 - (5)(17) - (5)(8) - (17)(8)}{18} = 6.5, \qquad \sigma_X = \sqrt{6.5} \approx 2.55.$$

The mean loading time is 10 minutes, with a standard deviation of about 2.55 minutes.

3.9 Checkout arrivals

Customers arrive independently at a checkout at a steady average rate of six customers per hour. Let X be the time in minutes between consecutive arrivals.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

A waiting time between independent events occurring at a steady rate identifies an exponential model. In minutes, $\lambda = 6/60 = 0.1$, so $X \sim \text{Exp}(0.1)$. Therefore,

$$E(X) = \frac{1}{0.1} = 10, \qquad Var(X) = \frac{1}{0.1^2} = 100, \qquad \sigma_X = 10.$$

The mean and standard deviation of the time between arrivals are both 10 minutes.

3.10 Machine breakdowns

Breakdowns of a machine occur independently at a steady average rate of four per hour. Let X be the time in hours from one breakdown to the next.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

The waiting-time clues identify $X \sim \text{Exp}(4)$. Hence,

$$E(X) = \frac{1}{4} = 0.25, \qquad Var(X) = \frac{1}{4^2} = 0.0625, \qquad \sigma_X = 0.25.$$

The mean time between breakdowns is 0.25 hour (15 minutes), and its standard deviation is also 0.25 hour.

3.11 Greenhouse temperature

Daily noon temperatures in a controlled greenhouse form an approximately symmetric, bell-shaped pattern centered at 68°F, with a standard deviation of 7°F. Let X be the temperature in degrees Fahrenheit.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

The symmetric bell shape identifies $X \sim N(68, 7^2)$. Thus,

$$E(X) = 68, \qquad Var(X) = 7^2 = 49, \qquad \sigma_X = 7.$$

The mean temperature is 68°F, and temperatures typically differ from it by about 7°F.

3.12 Package filling mass

A calibrated machine fills packages with masses that follow an approximately symmetric, bell-shaped pattern centered at 500 grams. The standard deviation is 25 grams. Let X be the mass of a package in grams.

Task: Identify the distribution and its parameters, then find $E(X)$, $Var(X)$, and σ_X.

Solution

The contextual shape identifies $X \sim N(500, 25^2)$. Therefore,

$$E(X) = 500, \qquad Var(X) = 25^2 = 625, \qquad \sigma_X = 25.$$

The mean filling mass is 500 grams, with a standard deviation of 25 grams.